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FMLCA: explainable and privacy-preserving federated machine learning classification algorithms for predicting heart disease in patients

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Abstract

Background Heart disease is a global health concern that significantly contributes to worldwide mortality. Machine Learning (ML) models have emerged as a powerful tool for predicting Coronary Artery Disease (CAD), a type of heart disease, by utilizing clinical features for classification. Federated Learning (FL) offers a solution for collaborative training without sharing raw data, thus addressing privacy concerns.

Methods This study presents an innovative approach, Federated Machine Learning Classification Algorithms (FMLCA), which utilizes cloud computing, privacy preservation techniques, and ML classification algorithms, including Decision Tree (DT), Adaptive Boosting (AdaBoost), K-Nearest Neighbors (KNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost), to predict CAD. In addition, privacy preserving is considered through the k-anonymity technique, and SHapley Additive exPlanations (SHAP) technique was utilized to identify features important in the model decision-making process.

Results The proposed RF model, compared to other models, obtained better performance. This RF model achieved an accuracy of 83.21% with privacy preservation and 84.49% without it. Furthermore, the SHAP technique enhances transparency by attributing feature influences in predictions.

Conclusion Implementing these models on a cloud platform results in efficient computational performance. This proposed approach represents a significant advancement in predictive healthcare tools, capable of accurately predicting CAD across distributed environments. By placing a strong emphasis on privacy and security, this approach underscores its importance and paves the way for a transformative healthcare ecosystem that centers on the needs of patients and healthcare providers.

Keywords Heart disease, Prediction, Federated learning, Privacy preservation, Machine learning

Background

Heart disease is a significant global health concern, including conditions such as Coronary Artery Disease (CAD), arrhythmia, and heart failure [1]. According to the World Health Organization (WHO), heart disease remains the leading cause of mortality worldwide, responsible for approximately 17.9 million deaths

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annually [2]. Timely and accurate detection of heart disease is essential for the effective management and prevention of life-threatening complications [1]. Recent global epidemiological studies further highlight the increasing burden of cardiovascular diseases (CVD), emphasizing the need for early risk stratification and advanced diagnostic tools [3, 4].

Machine Learning (ML) models have been developed as a promising auxiliary in healthcare, enabling predictive analytics and supporting clinical decision-making by analyzing complex datasets [5, 6]. ML algorithms process large and complex datasets, extracting valuable patterns and insights from various sources of patient information. In heart disease diagnosis, ML models can utilize clinical features such as blood pressure, cholesterol levels, glucose, age, gender, type of chest pain, electrocardiogram results, and exercise test results. Recent research demonstrates that ML-based prediction models outperform traditional statistical approaches by identifying non-linear relationships and subtle clinical patterns that are often overlooked in conventional assessments [7, 8].

ML algorithms like Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM) can classify clinical data into normal or abnormal categories. Deep learning models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) can learn complex patterns from clinical datasets [7]. Furthermore, deep learning has shown strong performance in cardiovascular diagnostics, including electrocardiogram interpretation, arrhythmia detection, and functional assessment of cardiac structure, supporting its growing clinical relevance [9, 10].

Sharing sensitive patient data between institutions poses significant challenges [11]. Data privacy, security, and regulatory compliance limit collaboration among healthcare providers [11]. Traditional centralization of data raises concerns over security breaches and unauthorized access [11, 12]. To overcome these barriers, Federated Learning (FL) allows multiple institutions to collaboratively train a shared model without exchanging raw patient data, thereby maintaining privacy [13–15]. Only model parameters are transmitted to a central server for aggregation, reducing the risk of data breaches and ensuring compliance with privacy regulations [16, 17]. Recent surveys and systematic studies highlight that FL has become a leading paradigm for collaborative healthcare modeling, specifically due to its capability to protect sensitive patient information across distributed clinical systems [18, 19].

Preserving patient privacy is paramount in healthcare, and some system achieves this through encrypted and anonymized data transmission from mobile devices to the central server [18, 20]. The use of encryption, differential privacy, and secure multiparty computation techniques ensures that patient data remains protected throughout the FL process [21, 22]. Advanced privacy-preserving mechanisms such as differential privacy, homomorphic encryption, and secure multiparty computation have been increasingly adopted in FL frameworks to mitigate model inversion and membership inference attacks [23, 24]. Moreover, the interpretability of ML models enhances trust among clinicians, providing insights into how models arrive at their decisions. The transparent framework bridges the gap between ML algorithms and clinical decision-making, empowering healthcare providers to confidently leverage the system's predictions in patient care. The current research emphasizes that integrating explainability into ML and FL systems is essential for clinical acceptance, as transparent models facilitate error analysis, bias detection, and medical decision support [25].

Combining privacy-preserving mechanisms with explainable ML forms the foundation of the proposed Federated Machine Learning Classification Algorithm (FMLCA). The FMLCA enables institutions to collaboratively develop predictive models while ensuring that sensitive patient data remains local. By integrating FL with model interpretability and privacy-enhancing techniques, the approach supports more accurate CAD as a type of heart disease prediction and facilitates clinically meaningful insights that may assist in personalized treatment planning.

The FMLCA allows healthcare institutions to utilize diverse datasets without directly exposing patient information to external servers or violating regulatory requirements. By addressing key challenges related to data privacy, security, and limited dataset sizes, this approach offers a practical and scalable path toward improving CAD diagnosis, particularly in environments where data sharing restrictions hinder the deployment of centralized machine learning systems. While previous studies have explored FL in healthcare [26–39], they have not fully integrated privacy-preserving mechanisms or provided a comprehensive interpretability component.

In contrast to existing FL studies that primarily focus on distributed model training without explicit privacy protection, this study introduces an integrated framework that incorporates k-anonymity and the IBM AI Privacy Toolkit to enhance data confidentiality. Moreover, the framework systematically evaluates multiple ML algorithms on a small clinical dataset and uses SHAP-based interpretability to identify and compare key predictive features. This combined use of FL, privacy-preserving techniques, and explainability fills a notable gap in the literature, as prior cardiac studies rarely address all three

aspects simultaneously [34]. Overall, the research gaps addressed in this study pertain to the following:

- Insufficient privacy preservation in FL-based cardiac prediction
- Limited interpretability of existing models
- Challenges associated with small and heterogeneous medical datasets
- Lack of integrated privacy-aware and explainable FL frameworks
- Limited discussion of practical implementation challenges in real healthcare settings

The main contributions of this study are summarized as follows:

1. A novel privacy-preserving and interpretable Federated Machine Learning Classification Algorithm (FMLCA) designed explicitly for coronary artery disease as a type of heart disease prediction.
2. Integration of privacy-preserving techniques (k-anonymity and IBM AI Privacy Toolkit) within the FL pipeline to enhance data confidentiality while maintaining predictive performance.
3. Comprehensive evaluation of multiple ML algorithms under both privacy-preserving and non-privacy-preserving FL settings using a small clinical dataset.
4. Incorporation of SHAP explainability to provide clinically relevant feature insights and improve model transparency.
5. A practical cloud-based implementation of the FMLCA, demonstrating feasibility in real-world, privacy-sensitive healthcare environments.

To the best of our knowledge, this is the first study to present a privacy-preserving, interpretable FMLCA for CAD prediction that combines FL, k-anonymity–based protection, and SHAP explainability within a unified system.

Related work

The integration of FL into healthcare, particularly for CVD prediction, has gained traction due to its ability to support collaborative model training across distributed datasets while preserving data privacy [35]. Traditional ML approaches typically require centralized data aggregation, which raises ethical and regulatory concerns under frameworks such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). In contrast, FL trains local models on decentralized data sources and aggregates only model parameters on a central server. Recent studies have proposed FL variants incorporating Graph Neural Networks (GNNs), optimization algorithms, and hybrid classifiers to improve predictive accuracy, computational efficiency, and privacy protection. This section reviews key studies, examining their methodologies and limitations, followed by a comparative synthesis highlighting gaps in scalability, multimodal integration, privacy, and real-world applicability, gaps addressed by the framework proposed in this work.

FL in healthcare

Several studies leverage biosignals or Electronic Health Records (EHRs) data within FL frameworks to predict CVD risk while maintaining data privacy.

Bhatt et al. [36] introduced an AI-based FL system for heart disease prediction that integrates region-based Convolutional Neural Networks (RCNN), Convolutional Neural Networks (CNNs), and Federated Logistic Regression (FLR). Using distributed datasets, their approach enables collaborative model development without centralizing sensitive data. Performance improved (up to 85% accuracy) compared with centralized ML approaches. However, synchronous model updates caused high communication overhead, and performance degraded under Non-Independent and Identically Distributed (Non-IID) data conditions, limiting applicability to heterogeneous clinical environments.

Ryu et al. [34] conducted a comparative review of 28 FL studies for CVD prediction, categorizing them into biosignal-based (17 studies) and EHRs-based (11 studies) approaches. Biosignal FL methods employed techniques such as client clustering and asynchronous learning to mitigate Non-IID challenges, whereas EHRs-based studies emphasized adaptive optimization and secure aggregation. While biosignal FL demonstrated high accuracy (≈95%) for personalized monitoring, limitations included insufficient cross-modality validation and unresolved interoperability constraints.

Alasmari et al. [37] proposed a multimodal FL framework with attention-based feature fusion and Stochastic Gradient Descent-Deep Neural Networks (SGD-DNN) for early CVD detection. Their system integrates cardiac images, electrocardiogram signals, EHRs, and nutrition data across distributed clients, achieving strong predictive performance (97.76–99.12%). Nonetheless, scalability remained a concern due to latency from large multimodal datasets, and the approach lacked explicit defenses against gradient leakage attacks.

Privacy-preserving in healthcare

Privacy-enhancing techniques, including differential privacy (DP), cryptography, and optimization-driven FL, are critical for clinical applications.

Khan et al. [30] developed an asynchronous FL approach (AFLCP) using deep neural networks (DNNs) for CVD prediction. By incorporating asynchronous updates and temporally weighted aggregation, their system mitigated class imbalance using federated SMOTE. While accuracy (98.43%) and communication efficiency improved, reliance on specific DNN architectures and limited experimentation on small datasets raised concerns regarding generalizability and real-world deployment.

Yaqoob et al. [28] presented a hybrid method combining Modified Artificial Bee Colony optimization with Support Vector Machines (MABC-SVM) for local feature selection and classification, combined with federated matched averaging on the server. Their method improved accuracy by 1.5% and reduced communication rounds by 17.7%. However, MABC-SVM's computational cost makes it unsuitable for resource-constrained devices, and robustness against adversarial attacks remains unaddressed.

Dash et al. [38] introduced FL with Weighted Conglomeration Optimization (FLWCO) for predicting diabetes and heart disease. Centralized pre-training followed by FLWCO yielded 97.27% accuracy and improved privacy compliance. Nevertheless, reliance on specific datasets (e.g., Pima Indian) and high pre-training costs limit general applicability, particularly in multimorbidity contexts.

Noor et al. [39] proposed a GNN-based FL framework for heart attack risk prediction, converting EHR data into graph structures distributed across three clients. Their approach achieved strong performance (96.79% accuracy; external validation 98.77%) but incurred considerable graph-construction overhead and demonstrated limited scalability for biosignal integration.

Edge computing in healthcare

Developing work integrates FL with edge and fog computing to reduce latency in CVD prediction and remote health monitoring.

Jeyaraman et al. [31] examined fog-based healthcare systems for real-time processing of biosignals and EHR data. Their findings highlight fog computing's ability to reduce latency in time-sensitive diagnostics such as stroke detection. However, challenges remain in interoperability, scalability under high data volumes, and exposure to security threats due to the absence of FL-specific privacy mechanisms.

Li et al. [32] developed ADDETECTOR, an FL-enabled innovative healthcare system for Alzheimer's disease detection, extendable to CVD monitoring. Using Internet of Things (IoT) audio data and a three-layer architecture (user-client-cloud), their approach integrates DP and cryptographic aggregation while maintaining low computational overhead (0.7s). Despite this, its reliance on audio-derived features limits generalizability, and cryptographic aggregation introduces additional system complexity.

Summary and identified research gaps

The reviewed studies differ substantially in data modality, model architecture, and aggregation strategies. Biosignal-focused methods [33, 34, 37] prioritize real-time processing via attention mechanisms or IoT integration, offering highly personalized predictions but remaining sensitive to Non-IID data, a limitation partially mitigated through clustering [34] or asynchronous updates [30]. EHRs-based approaches [29, 38, 39] emphasize scalability and optimized aggregation, achieving robust accuracy (95–99%) but rarely incorporating multimodal biosignals. Hybrid methods [29, 30] blend classifiers (e.g., MABC-SVM, AdaBoost) with FL, reducing communication rounds by 17–70% [29, 35], while GNN-enhanced FL [39] excels in graph-structured data handling (96.79% accuracy) but incurs higher computational costs. Privacy mechanisms vary: DP and cryptography are common [32, 36], yet asynchronous FL [30] and weighted optimization [38] better handle efficiency, with trade-offs in convergence speed.

Findings reveal consistent accuracy gains over centralized baselines (1.5–2% improvements [29, 38]), with FL enabling privacy-compliant collaboration across hospitals or IoT devices. Multimodal frameworks [37] demonstrate superior generalizability (99.12% on diverse datasets), while fog/edge integrations [31] reduce latency for urgent diagnostics. However, limitations are prevalent; most studies [30, 32, 34, 36] evaluate on small or synthetic datasets, risking overfitting and poor real-world robustness. Communication overhead remains high in synchronous FL [36, 37], and security vulnerabilities (e.g., gradient inference attacks) are underexplored [29, 39]. Scalability issues arise in multimodal [37] and edge [31] contexts, with Non-IID data causing 5–10% accuracy drops [30, 34]. Comorbidity handling (diabetes-CVD [38]) is limited, and interpretability crucial for clinical adoption is rarely addressed.

Taken together, existing FL approaches have advanced privacy-preserving CVD prediction but continue to face challenges in scalable multimodal integration, robustness under heterogeneous data conditions, defense against advanced privacy attacks, and deployment in low-latency, resource-constrained environments. This study addresses these limitations by introducing an enhanced FL framework that improves predictive accuracy, supports scalable multimodal data integration, strengthens privacy protection, and increases robustness and computational

efficiency across distributed and heterogeneous healthcare settings.

Materials and methods

The steps taken in this study are described as follows:

Preprocessing and feature analysis:

The dataset initiates the process with a thorough preprocessing phase aimed at enhancing its quality and suitability for analysis. After the relevancy analysis, the dataset undergoes normalization to standardize the data distribution, ensuring consistency in subsequent processing steps. Subsequently, the interrelationships between dataset features undergo assessment using the Pearson correlation coefficient. This initial phase ensures the dataset is optimized for subsequent analysis.

Central and local models creation:

The methodology involves the creation of five local models. Within each local model, the proposed ML algorithms are initiated and trained in a federated form. These local models establish a secure connection with a central model, facilitating the sharing of model weights among them.

Evaluation and privacy measures:

The performance of the FMLCA in predicting CAD is rigorously evaluated. This evaluation encompasses a comprehensive set of metrics, ensuring a thorough assessment of the model's effectiveness. Moreover, privacy protection measures are integrated to safeguard sensitive information during the federated training process. Also, a statistical test was conducted to assess the significance of the models' performance.

Interpretability analysis:

Concluding the approach, the SHapley Additive exPlanations (SHAP) method is applied to discern the most influential features guiding the decision-making processes of the ML models. This step provides valuable insights into the factors contributing significantly to the model's predictions.

Dataset

The dataset utilized in this study consists of 303 samples and includes 13 distinct features with a binary class label (0 or 1), focusing on predicting CAD as a type of heart disease in patients [33]. Additional details about the dataset are provided in Table 1.

Dataset preprocessing

The dataset utilized in this study underwent a meticulous preprocessing phase to ensure its suitability for subsequent ML modeling. Initially, the dataset contained instances with missing values, which could introduce errors or bias during model training. As a precaution, 18 samples with missing values in essential clinical variables were removed. This approach was preferred over imputation because, in a small clinical dataset, imputing non-random missing values can distort feature relationships and affect interpretability methods such as SHAP.

After this cleansing process, the dataset comprised 284 complete samples, divided into two classes: Class 1, representing the presence of heart disease, included 126 samples, and Class 0, representing the absence of the disease, included 158 samples. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively on the cleaned training dataset. This procedure generated synthetic samples for

Table 1 The dataset description

Feature	Description	Mean	Standard deviation
Age	Age	54.4	9
Sex	Sex	N/A	N/A
cp	Chest pain type	N/A	N/A
trestbps	Resting blood pressure	131.6	17.5
chol	Serum cholesterol	246.3	51
fbs	Fasting blood sugar	N/A	N/A
restecg	Resting electrocardiographic results	N/A	N/A
thalach	Maximum heart rate achieved	149.6	22.9
exang	Exercise-induced angina	N/A	N/A
oldpeak	ST depression induced by exercise relative to rest	1.04	1.16
Slope	The slope of the peak exercise ST segment	N/A	N/A
ca	Number of major vessels (0–3) colored by fluoroscopy	0.67	0.94
thal	Thallium heart scan	N/A	N/A

the minority class, resulting in a balanced dataset of 316 samples, with each class comprising 158 samples.

This preprocessing strategy ensured that the dataset was both complete and balanced, providing a reliable foundation for training robust ML models while maintaining data integrity and interpretability.

In addition, the dataset underwent normalization using the min–max normalization and standard scaler techniques [40]. Min–max normalization harmonized the range of values across the dataset, making it suitable for subsequent ML modeling. Each feature's values were transformed into a range between 0 and 1, following Eq. (1).

In this formula, X_{max} is the maximum value and X_{min} is the minimum value of each feature.

$$X_{normalize} = \frac{X - X_{min}}{X_{max} - X_{min}}. \quad (1)$$

The standard scaler method standardizes features by removing the mean and scaling to unit variance. Each feature X is transformed according to Eq. (2):

$$X_{Standard} = \frac{x - \mu}{\sigma}, \quad (2)$$

where μ is the mean and σ is the standard deviation of the feature. This transformation ensures that features have zero mean and a standard deviation of one, which can improve the convergence and stability of algorithms.

Through the diligent implementation of these preprocessing measures, the dataset underwent careful refinement. This meticulous process is intended to facilitate the development of robust and accurate ML models, ultimately enabling the provision of reliable insights into the prediction of CAD.

Feature investigation

The Pearson correlation coefficient measures the strength and direction of a linear relationship between two continuous variables, ranging from −1 (perfect negative correlation) to + 1 (perfect positive correlation), with 0 indicating no linear correlation.

The SHAP method is a widely recognized technique within the realm of eXplainable Artificial Intelligence (XAI) [41]. Its primary purpose is to enhance our understanding of ML model predictions. This is achieved by deconstructing individual predictions and attributing the impact of each feature to the model's output [41]. The foundation of the SHAP method lies in SHAP values, derived from principles of cooperative game theory, such as the shapley value [41]. Essentially, the SHAP method offers a nuanced and consistent approach to unraveling the predictions of ML models. It considers feature interactions, providing precise insight into the elements influencing individual predictions. Through the assignment of informative weights to each feature, SHAP plays a pivotal role in enhancing the clarity, accountability, and comprehensibility of AI and ML models [41]. This transparency is paramount in instilling confidence in decision-making driven by AI [41].

Privacy-preserving technique

K-anonymity serves as a pivotal privacy-preserving strategy and technique dedicated to the protection of individuals' sensitive data contained within datasets. It operates on the fundamental principle that, in data releases for analysis, each record must remain indiscernible from a minimum of "k" other records, particularly concerning specific identifying attributes. This anonymization mechanism effectively thwarts any potential re-identification of individuals from the disseminated data, all the while preserving the dataset's analytical value intact [42].

The crux of achieving k-anonymity lies in the manipulation of quasi-identifiers, a set of attributes that, when combined, pose a risk of enabling re-identification. Through techniques such as generalization or suppression, the granularity of the dataset is deliberately curtailed, rendering individual records completely unidentifiable [42].

The pursuit of k-anonymity necessitates striking a delicate balance between privacy and data utility. While aggressive generalization or suppression may enhance privacy, it can potentially hinder the dataset's analytical utility [42]. In the context of applying k-anonymity to datasets intended for ML, the preprocessing phase is executed before model training. This critical step ensures the protection of confidential data, with models being refined based on generalized or suppressed data. Subsequently, ML algorithms are deployed on the transformed dataset for in-depth analysis and predictive modeling [43, 44].

ML models

In this study, various ML models were considered to ensure a comprehensive evaluation of predictive performance and suitability for small medical datasets. Specifically, Random Forest (RF), Decision Tree (DT), AdaBoost, XGBoost, and K-Nearest Neighbors (KNN) were selected as representative and widely used algorithms in clinical research. These models are commonly applied in medical dataset analysis, particularly for small datasets, as discussed in the previous review [45]. This selection allows for transparent comparison among different modeling approaches, ensuring that the final model is empirically justified based on accuracy, robustness, and interpretability. By assessing multiple standard models, we aimed to identify the most effective algorithm for integration into the proposed FMLCA, thereby

supporting its potential application as a reliable clinical decision support tool in hospitals and healthcare settings.

The DT algorithm is a widely used method in ML, adept at solving classification and regression tasks. It constructs a tree-like structure to navigate complex decision scenarios and their potential outcomes. The core idea is to simplify intricate decisions into a sequence of simpler ones, making it easily interpretable and visually understandable. DTs stand out due to their simplicity, interpretability, and versatility in handling varied features [46].

The splitting criterion at each node can be defined using the Gini:

$$Gini(D) = 1 - \sum_{i=1}^{C} p_i^2,$$

where p_i is the proportion of class i in dataset D and C is the total number of classes. Hyperparameters for this model include maximum depth, minimum samples per leaf, and the splitting criterion.

The RF algorithm stands as a potent ensemble learning technique extensively harnessed in ML for tackling both classification and regression tasks. This method is an evolutionary extension of DTs, engineered to enhance their performance by mitigating overfitting and elevating predictive precision. The cornerstone of an RF comprises an ensemble, or collection, of distinct DTs. Each tree emerges from a unique subset of the training data and, potentially, a portion of the available features [46].

The ensemble prediction is performed using majority voting:

$$\widehat{y} = mode\{h_1(x), h_2(x), \ldots, h_n(x)\},$$

where $h_i(x)$ is the prediction of the i-th tree and n is the total number of trees. This model's hyperparameters include the number of trees, maximum depth, and minimum samples per leaf.

The introduction of diversity courtesy of random data and feature subsets, coupled with the ensemble nature of the RF, serves as a countermeasure against overfitting. This synergy culminates in a more stable, reliable model poised to handle new and unseen data adeptly. Moreover, the RF algorithm harnesses the combined wisdom of multiple DTs, each honed on a distinct data and feature subset. This synthesis yields a comprehensive, exacting model that surmounts the limitations inherent to individual DTs, rendering it a premier choice across a spectrum of ML applications [46].

The Adaptive Boosting (AdaBoost) algorithm serves as a versatile ensemble learning method applicable to both classification and regression tasks. Its core objective lies in refining the performance of weak learners (simple models just slightly better than random guessing) by intelligently amalgamating their forecasts. AdaBoost operates by rectifying the errors made by previous ensemble members, making it a robust strategy for bolstering predictive accuracy across various problem types [47].

This model's final prediction is

$$\widehat{y} = sign\left(\sum_{m=1}^{M} \alpha_m h_m(x)\right),$$

where M is the number of weak learners, h_m is the prediction of the m-th weak learner, and α_m is its assigned weight based on performance.

The Extreme Gradient Boosting (XGBoost) algorithm stands as a sophisticated and remarkably efficient ensemble learning technique, acclaimed for its exceptional performance in both ML competitions and real-world scenarios. It represents an evolved rendition of the gradient boosting algorithm, engineered to tackle a broad spectrum of tasks encompassing regression, classification, ranking, and tailored prediction challenges. XGBoost's hallmark traits encompass rapid computation, precision, and the capacity to manage substantial datasets [48].

The objective function of this mode is

$$Obj = \sum_{i} l(y_i, \widehat{y_i^{(t)}}) + \sum_{k} \Omega(f_k),$$

where l is the loss function, f_k is the k-th tree, and $\Omega(f_k)$ is the regularization term. For this model, hyperparameters include learning rate, maximum depth, and number of estimators.

The KNN algorithm stands as a fundamental yet potent technique in ML, applied for tasks of classification and regression alike. Operating as a non-parametric, instance-based learning method, it forms predictions for new data points by gauging their proximity to existing data points within a feature space. Predictions are based on the majority class of the K-nearest neighbors:

$$\widehat{y} = mode\{y_i | x_i \in Knearestneighbors\}.$$

The bedrock concept driving KNN is that similar instances tend to yield comparable outcomes [46]. A pivotal role is played by the distance metric, serving as the yardstick for measuring likeness between instances. Commonly, the Euclidean distance or Manhattan distance is used:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{p} (x_j - x_{ij})^2},$$

where p is the number of features.

The calculated distance between two instances encapsulates their degree of resemblance or divergence [46]. A critical decision entails the choice of the parameter "K," signifying the count of nearest neighbors to consider during prediction. This parameter, a hyperparameter, requires predetermination before algorithm application [46].

Model implementation

Figure 1 shows the workflow diagram of the developed models in this study. In the implementation of the models, five local models were created using the Docker tool. The dataset was initially divided into five folds, and each of these folds was assigned to a specific local model. Within each local model, a five-fold cross-validation methodology was applied, where each fold served as the testing set once, and the remaining folds were used for training. This study pursued two distinct approaches.

Additionally, it is noteworthy that five iterations for updating local models and the general model were employed. This iterative process, combined with the five-fold cross-validation, enhances the learning and adaptation of the models over five cycles, contributing to the refinement of the overall Federated Machine Learning (FML) approach (global models). The use of multiple iterations allows the models to progressively improve their performance by updating based on insights gained from each round of training and evaluation within specific folds of the dataset.

Regarding the aggregation method employed in this study, the local models transmitted their weights to the central server five times during the iterative process. At the central server, the aggregation of these weights was performed using an averaging approach. Specifically, the weights of the five locally trained RF models were averaged to generate the global model. This method ensured that the central model effectively integrated the knowledge learned from each local model, preserving the decentralized structure of the FL.

Furthermore, the weights of the global model were sent back to the local models after each aggregation step, completing an iterative feedback loop. These updated

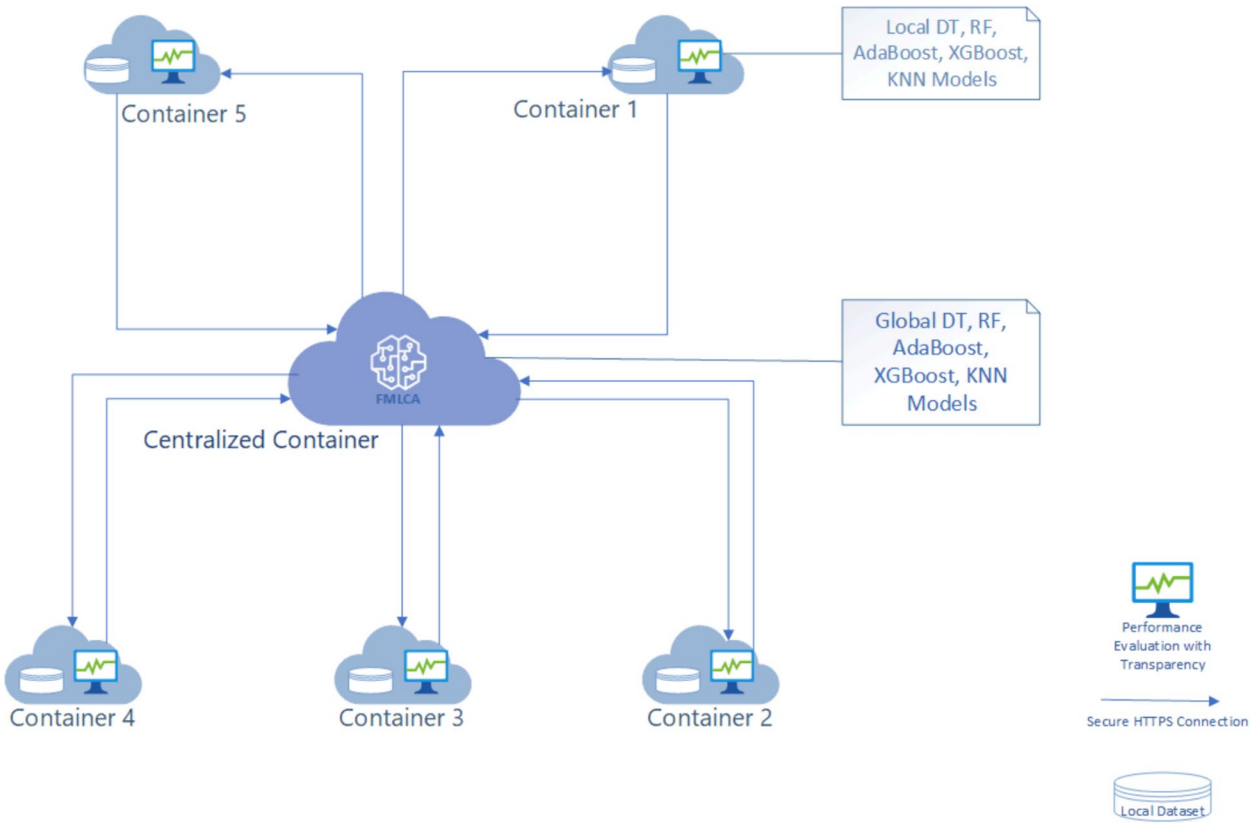


Fig. 1 Structure of FMLCA models

global weights served as the initial starting point for the subsequent training round in each local model. This bidirectional exchange of weights between the local models and the central server occurred five times, ensuring that both the local models and the general model progressively adapted and improved over the iterations.

The iterative updates and the bidirectional communication between the local and global models significantly enhanced the general model's performance and robustness. By leveraging this iterative approach, the FL process effectively captured the diverse patterns present in the distributed dataset while ensuring continual refinement and optimization of the global model.

First approach (with privacy preservation)

In this approach, the development of the proposed ML models involved the utilization of the IBM AI Privacy Toolkit [44] library in the Python programming language. This toolkit incorporates an anonymization module that plays a crucial role in preserving privacy while predicting CAD in patients. A fundamental component of this module is the application of the k-anonymity technique, which was employed on the dataset before it was divided into one-fifths for each local model. In this approach, the development of the proposed ML models involved the utilization of the IBM AI Privacy Toolkit [44] library in Python. This toolkit provides a comprehensive suite of privacy-preserving functionalities designed to ensure data confidentiality and regulatory compliance during model training. A key feature of the toolkit is its anonymization module, which was employed to protect sensitive patient information before splitting the dataset into five subsets for the local models. Specifically, the k-anonymity technique was applied to the dataset to prevent re-identification of individuals, ensuring that each record could not be distinguished from at least $k-1$ other records with respect to the quasi-identifiers. The toolkit also allows for the configuration of privacy parameters, including the selection of optimal k values, which were determined through grid search alongside hyperparameter tuning for the ML algorithms.

The IBM AI Privacy Toolkit additionally provides security assurances by enforcing strict privacy guarantees throughout the federated learning process. These assurances include the following:

- Maintaining data anonymization during local processing.
- Preserving confidentiality when transmitting model parameters to the central server.
- Integrating seamlessly with FL frameworks to avoid exposing raw data to external networks.
- Allowing iterative privacy-compliant updates in multi-round FL training.

By using these functionalities, the toolkit ensures that the first approach maintains robust privacy protection while allowing the federated models to learn from distributed datasets collaboratively. This integration highlights the toolkit's role in balancing privacy preservation with predictive performance, enabling practical clinical applicability.

Second approach (without privacy preservation)

In this alternative approach, the ML models were developed without the incorporation of privacy-preserving measures. The primary objective remained the prediction of CAD in patients; however, privacy considerations were not a focal point in this instance.

All five local models established a connection with a central counterpart, enabling the iterative updating of their weights. This dynamic interaction culminated in the collective prediction of CAD in patients through the FL.

To optimize the models' performance and minimize potential learning errors, the grid search methodology was adopted. This strategy facilitated the identification of the most suitable hyperparameters for the ML algorithms. By applying this search technique, the optimal values for hyperparameters, as well as the optimal k value in the k-anonymity technique, were determined. Table 2 and Table 3 outline the selected hyperparameters and k values for the first and second approaches, respectively.

Table 2 Best ML algorithms' hyperparameters with an anonymity approach

Model	Hyperparameters
DT	spliter: best, min_samples_split: 2, criterion: gini, max_depth: none
RF	n_estimators: 100, criterion: gini, estimator_min_samples_split: 2, max_depth: none
AdaBoost	n_estimators: 100, algorithm: SAMME.R, learning_rate: 0.1
XGBoost	sampling method: gradiant_based, n_estimators = 100, eta: 0.3, booster: gbtree
KNN	algorithm: kd_tree, n_neighbors: 3, p: 2
K (k-anonymity)	100

Table 3 Best ML algorithms' hyperparameters without an anonymity approach

Model	Hyperparameters
DT	spliter: best, min_samples_split: 2, criterion: gini, max_depth: none
RF	n_estimators: 100, criterion: gini, estimator_min_samples_split: 2, max_depth: none
AdaBoost	n_estimators: 50, algorithm: SAMME.R, learning_rate: 0.1
XGBoost	sampling method: gradiant_based, n_estimators = 100, eta: 0.1, booster: gbtree
KNN	algorithm: kd_tree, n_neighbors: 5, p: 2

Model evaluation

Following the determination of optimal hyperparameters, the proposed ML algorithms underwent training within the FL. The effectiveness of these models was gauged through the utilization of the AUC (Area Under the Curve) metric [49]. In addition, a comprehensive assessment encompassed six other criteria, namely accuracy, precision (Positive Predictive Value or PPV), Negative Predictive Value (NPV), specificity, sensitivity, or recall, and F1-Score, all of which were employed to evaluate the models [50] rigorously.

The evaluation criteria were calculated according to the following equations [50]:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}, \tag{3}$$

$$Precision or Positive Predictive Value(PPV) = \frac{TP}{FP + TP}, \tag{4}$$

$$Negative Predictive Value(NPV) = \frac{TN}{FN + TN}, \tag{5}$$

$$Specificity = \frac{TN}{FP + TN}, \tag{6}$$

$$Sensitivity or Recall = \frac{TP}{FN + TP}, \tag{7}$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}. \tag{8}$$

In the context of Eqs. (3–8), TP stands for True Positive, FP stands for False Positive, TN stands for True Negative, and FN stands for False Negative.

Furthermore, to assess the significance of performance differences among the implemented models, pairwise sample t-test were conducted on the primary evaluation metric across the local models in both FL approaches [51]. This statistical analysis helps determine whether the observed differences in model performance are statistically significant ($p < 0.05$), providing additional confidence in the robustness and reliability of the results.

These comprehensive metrics provide a holistic understanding of the models' performance, enabling a practical assessment of their predictive capabilities.

Results

This study involved the implementation of various ML algorithms within the approach of FL, followed by an evaluation of their efficacy using predefined evaluation criteria across two distinct approaches. The interrelation between dataset features was depicted through Pearson's correlation coefficient, as shown in Fig. 2, revealing that no feature exhibited a strong correlation exceeding 0.5 with any other feature. This indicates that multicollinearity among features is minimal, suggesting that each feature provides unique information to the models and that the observed predictive performance is unlikely to be biased due to redundant features.

Although the min–max method demonstrated superior performance across the models, the standard scaler method was also evaluated, and its comparative results are presented in Table A1 and Table A2 in Appendix A for both approaches of this study. Through the careful implementation of these preprocessing steps, the dataset was systematically refined to support the development of robust and accurate ML models capable of providing reliable insights into CAD prediction.

The outcomes of the models in the first approach are extensively detailed in Table 4, showcasing evaluation results based on diverse metrics. Notably, the RF algorithm demonstrated superior performance across all evaluation criteria among all proposed models. The ensemble learning structure of the RF algorithm, where the collective decision of multiple trees contributes to its overall high performance, played a fundamental role in these promising results [46]. Remarkably, AdaBoost and XGBoost performed well in some local models, but their average F1-Score and accuracy remained lower than RF, emphasizing the reliability of RF as the final choice.

In Table 5, the computational time dedicated to each model within individual local models in the first approach

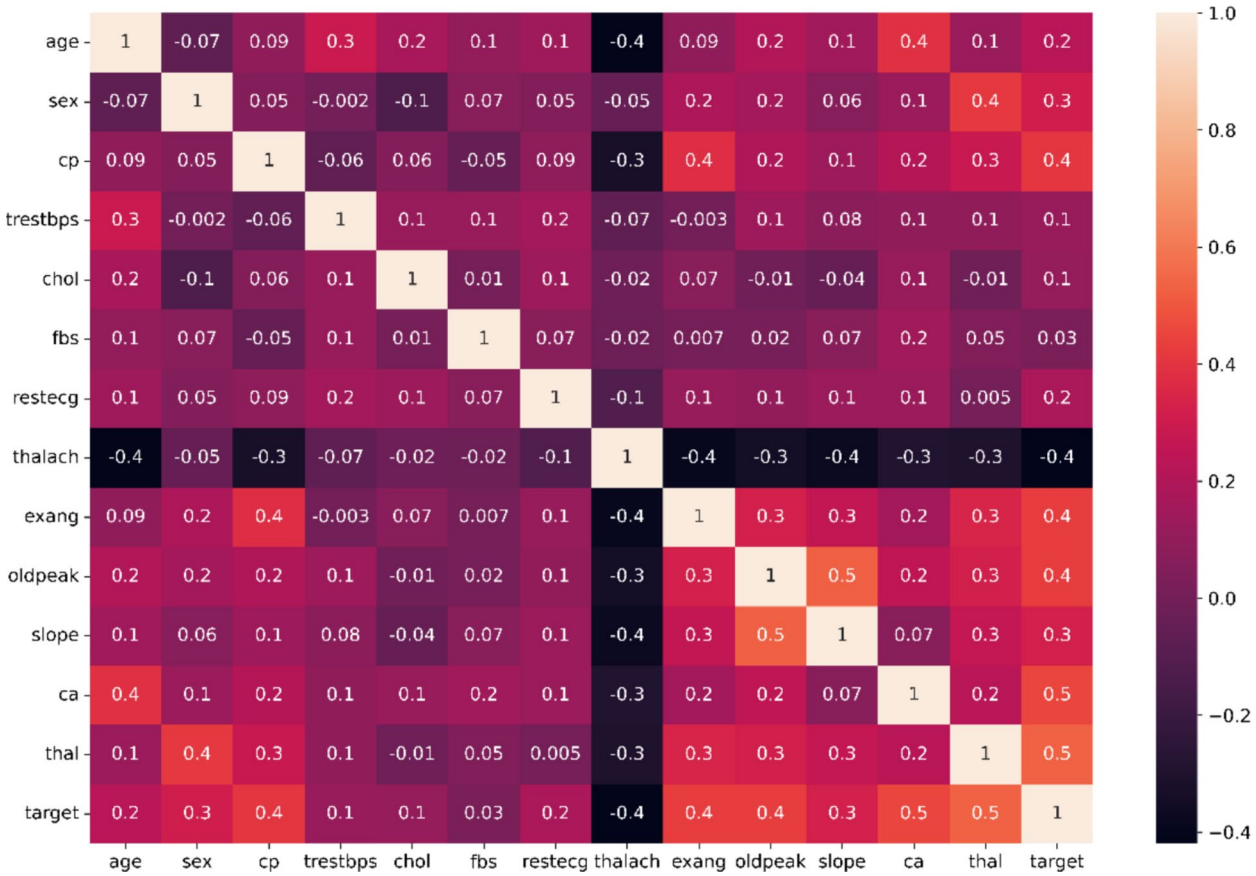


Fig. 2 A visual representation of Pearson correlation coefficient values between features

is outlined. It is important to clarify that the computational time was measured through multiple runs, and the reported values represent the averaged results. This approach enhances the reliability and accuracy of the reported computational time. A more graphical representation of the RF model's performance in terms of AUC is presented in Fig. 3. Using the SHAP technique, Fig. 4 and Fig. 5 illustrate the primary features influencing the first approach of the RF model.

Transitioning to the second approach, Table 6 meticulously displays the performance outcomes of the models, once again highlighting the consistent superiority of the RF algorithm across various evaluation measures between all proposed models. Complementary to this, Table 7 provides insights into the computational time allocated to each model per local model in the second approach. Visualizing the RF model's AUC performance in the second approach, Fig. 6 captures the essence of this aspect. The SHAP technique once more proved invaluable in unraveling the critical features shaping the RF model's second approach, which are graphically presented in Fig. 7 and Fig. 8.

Nonetheless, according to Table 5 and Table 7, while RF achieved superior predictive performance, it incurred moderately higher computational time compared to simpler models such as DT and KNN. This trade-off is acceptable in clinical contexts where accuracy and interpretability are prioritized over minimal processing time. The slightly increased time in the first approach also reflects the overhead introduced by privacy-preserving measures like k-anonymity.

SHAP analysis revealed the most influential in predicting CAD across both approaches. These findings are consistent with clinical knowledge, where these features are critical indicators of heart disease. Subtle differences in the ranking of features between the two approaches may be attributed to the privacy-preserving modifications in the first approach, which slightly affected feature contributions. The high AUC values in both approaches indicate that the RF model has strong discriminative power, reliably distinguishing between patients with and without CAD.

Additionally, to validate the observed differences in model performance, sample t-test was conducted on the average accuracy of all local models in both FL

Table 4 Performance of models in the first approach

Model	Container	Accuracy (%)	PPV (%)	NPV (%)	Specificity (%)	Sensitivity or Recall (%)	F1-Score (%)
DT	Local model 1	76.56	77.14	75.86	73.33	79.41	78.26
	Local model 2	76.19	82.14	71.42	83.33	69.69	75.41
	Local model 3	71.42	63.63	80.00	66.66	77.77	70.00
	Local model 4	74.60	75.67	73.07	67.85	80.00	77.77
	Local model 5	76.19	76.92	75.67	82.35	68.96	72.72
	Average	74.99	75.10	75.20	74.70	75.17	74.83
RF	Local model 1	85.93	85.71	86.20	83.33	88.23	86.95
	Local model 2	80.95	74.19	87.50	77.77	85.18	79.31
	Local model 3	80.95	83.87	78.12	83.33	78.78	81.25
	Local model 4	87.30	84.61	91.66	78.57	94.28	89.18
	Local model 5	80.95	81.48	80.55	85.29	75.86	78.57
	Average	**83.21**	**81.97**	**84.81**	**81.66**	**84.41**	**83.05**
AdaBoost	Local model 1	89.06	88.57	89.65	86.66	91.17	89.85
	Local model 2	77.77	85.18	72.22	86.66	69.69	76.66
	Local model 3	85.71	80.95	95.23	71.42	97.14	88.31
	Local model 4	79.36	71.87	87.09	75.00	85.18	77.96
	Local model 5	74.60	72.41	76.47	76.47	72.41	72.41
	Average	81.30	79.79	84.13	79.24	83.12	81.04
XGBoost	Local model 1	90.62	91.17	90.00	90.00	91.17	91.17
	Local model 2	80.95	74.19	87.50	77.77	85.18	79.31
	Local model 3	82.54	86.66	78.78	86.66	78.78	82.54
	Local model 4	79.36	78.94	80.00	71.42	85.71	82.19
	Local model 5	77.77	74.19	81.25	76.47	79.31	76.66
	Average	82.25	81.03	83.50	80.46	84.03	82.37
KNN	Local model 1	79.68	81.81	77.41	80.00	79.41	80.59
	Local model 2	74.60	66.66	83.33	69.44	81.48	73.33
	Local model 3	80.95	86.20	76.47	86.66	75.75	80.64
	Local model 4	87.30	88.57	85.71	85.71	88.57	88.57
	Local model 5	73.01	66.66	81.48	64.70	82.75	73.84
	Average	79.11	77.98	80.88	77.30	81.59	79.39

approaches. The t-test results confirm that the superior performance of the RF model is statistically significant ($p-\text{value} < 0.05$) compared to the other algorithms. This supports the robustness of RF in this clinical dataset and justifies its selection as the optimal model for CAD prediction. The consistent superiority of RF across multiple metrics and local models demonstrates its suitability for developing reliable, clinically applicable predictive tools Tables 8 and 9.

Discussion

In this study, we embarked on the development of five FMLCAs, utilizing a cloud platform orchestrated by Docker, to predict heart disease, CAD, in patients. In this study, two distinct approaches were considered to compare FL models with and without privacy preservation, by also using an interpretable method for transparency. Across both approaches, the RF algorithm emerged as a standout performer, showcasing promising results. The statistical significance test (t-test) conducted on model performances further confirmed that the superior performance of the RF model was not due to random variation, supporting its robustness and clinical relevance. The selection and interpretation of evaluation metrics have direct clinical significance in predicting CAD. Depending on the clinical context, different metrics may be prioritized: in a screening scenario, maximizing sensitivity (recall) helps identify most potential disease cases and minimize false negatives, which is critical to avoid missing at-risk patients; in a diagnostic scenario, precision is important to reduce false positives and avoid unnecessary interventions or patient anxiety. Metrics such as

Table 5 Computational time of models in the first approach

Model	Local model	Time (second)
DT	Local model 1	1.18
	Local model 2	1.52
	Local model 3	2.83
	Local model 4	3.42
	Local model 5	2.71
	Average	2.33
RF	Local model 1	3.11
	Local model 2	3.17
	Local model 3	4.14
	Local model 4	4.32
	Local model 5	4.12
	Average	3.77
AdaBoost	Local model 1	4.95
	Local model 2	5.26
	Local model 3	5.21
	Local model 4	5.17
	Local model 5	9.95
	Average	6.10
XGBoost	Local model 1	9.01
	Local model 2	6.98
	Local model 3	7.40
	Local model 4	7.37
	Local model 5	14.83
	Average	9.11
KNN	Local model 1	3.55
	Local model 2	3.37
	Local model 3	3.37
	Local model 4	3.50
	Local model 5	3.55
	Average	3.46

F1-Score and AUC offer a balanced and comprehensive assessment of model performance, capturing both the trade-offs between sensitivity and specificity and the model's ability to discriminate between disease and non-disease across thresholds. By considering both F1-Score and AUC, we can assess not only the model's classification performance at a chosen threshold but also its overall discriminative power, a combination beneficial for medical datasets, where both classes' outcomes have clinical importance. This dual-metric evaluation provides a more reliable and clinically meaningful measure of performance than accuracy alone. As shown in recent literature, F1-Score and AUC remain among the most recommended metrics for evaluating ML models on medical datasets because they provide complementary insights into classification performance and discriminative ability [52, 53]. Because our dataset, after balancing with SMOTE, aims to approximate equal class representation, employing both F1-score and AUC allows us to verify that our chosen model (RF) maintains both high discriminative capacity and robust classification performance in clinically relevant terms, independent of threshold bias.

Nonetheless, in the first approach, the RF model achieved an average accuracy of 83.21% and an average F1-Score of 83.05%, while in the second approach, it attained an accuracy of 84.49% and an F1-Score of 86.13%. Notably, the incorporation of the k-anonymity technique in the first approach led to a relatively decreased efficiency compared to the second approach. Additionally, the RF model in the first approach incurred an extra 3.653 s in computing time compared to the second approach. Moreover, the RF algorithm garnered average AUC values of 89.66% and 92.16% in the first and second approaches, respectively. As detailed in [49], the performance of the proposed RF model was deemed acceptable and outstanding based on the AUC metric in the first and second approaches, respectively.

This study addresses key gaps in FML model research in healthcare. It implements and evaluates FL models on a cloud platform with privacy-preserving techniques for CAD prediction, ensuring both data confidentiality and model applicability. By using a patient-balanced dataset, the models achieve more reliable and unbiased predictions. Optimal hyperparameter tuning was performed to enhance model efficiency and robustness. Additionally, the application of an interpretable technique to identify the most influential features provides insights into the RF model's decision-making process and supports clinical trust in its predictions.

Comparatively, earlier research primarily focused on the implementation of ML algorithms within cloud platforms [26]. For instance, Desai et al. [26] introduced an ensemble learning approach, with the Support Vector Classifier model as the best model, achieving an accuracy of 82.46% and a mere 0.016 s in computing time on a test dataset. Nevertheless, our developed RF model achieved a commendable accuracy of 83.21% and a computing time of 3.77 s on average, even while factoring in privacy preservation. Given the paramount significance of patient data confidentiality within healthcare systems globally, a slight computational delay of this magnitude remains justifiable [54]. Moreover, previous endeavors in the use of privacy-preserving FL for healthcare applications, like diagnosing AD, incurred substantial computational overhead to ensure data privacy. This work has increased the computing time by 1.56% seconds per sample [32].

The implementation of FMLCA, integrated with the IBM AI Privacy Toolkit, marks a significant stride in ensuring privacy preservation within the FL. This approach achieves the crucial objective of maintaining

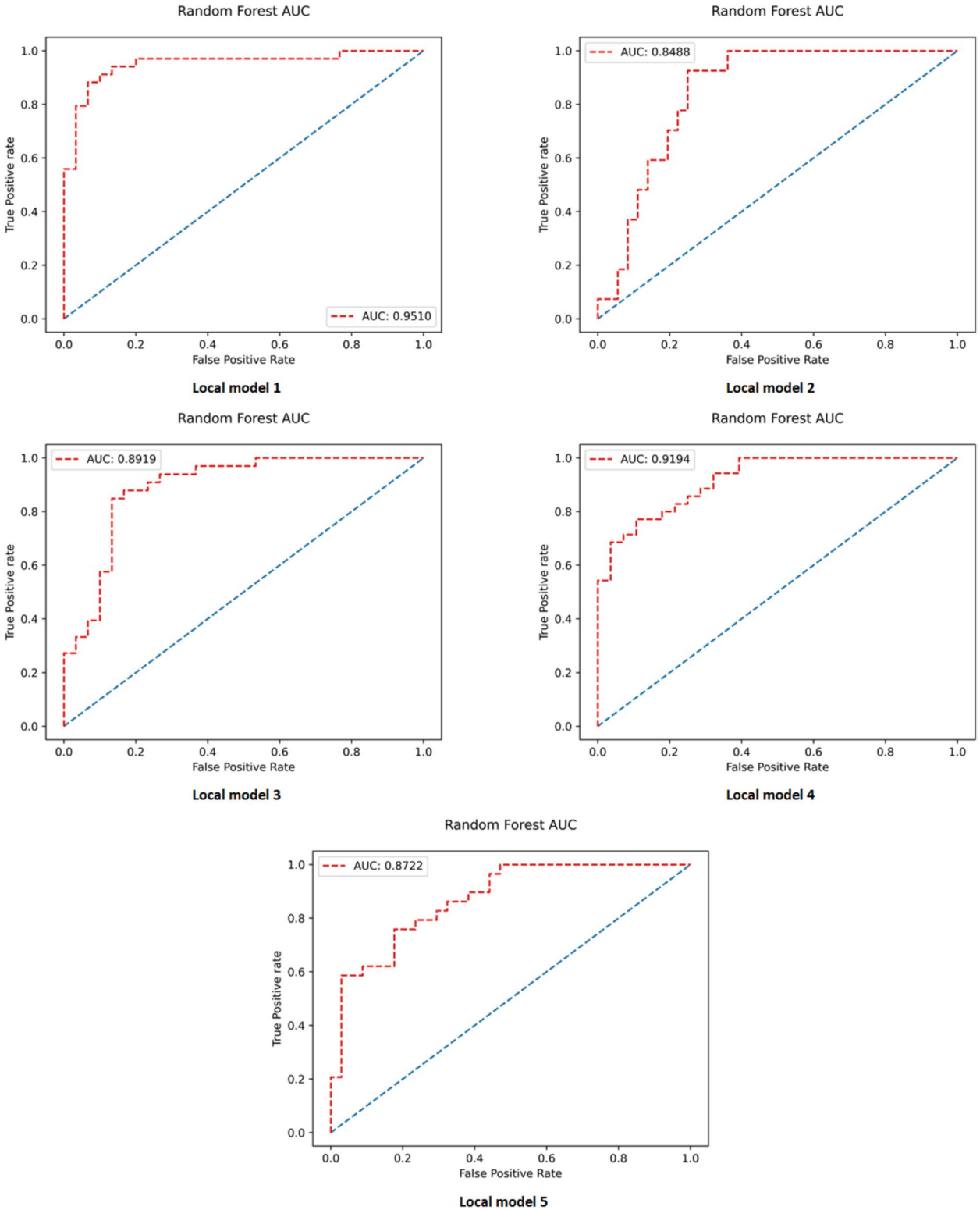


Fig. 3 AUC diagrams depicting the RF model's performance in the first approach

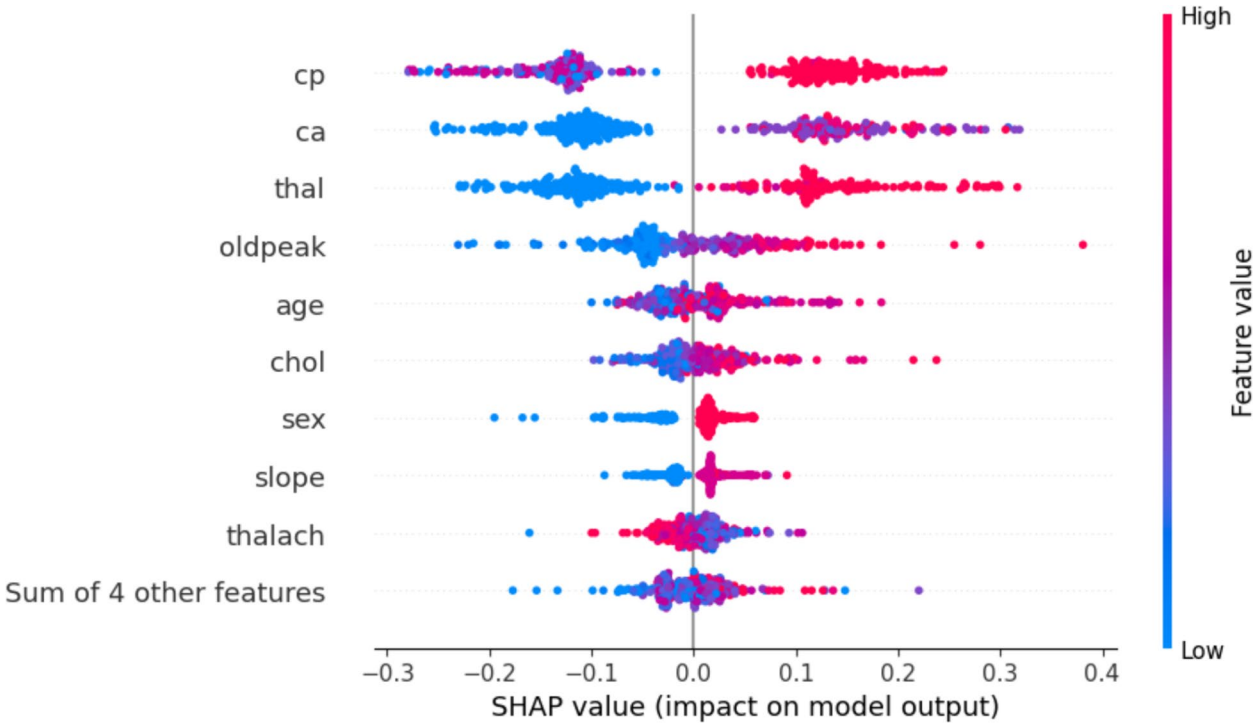


Fig. 4 SHAP values depicting the influence of features on the RF model in the first approach

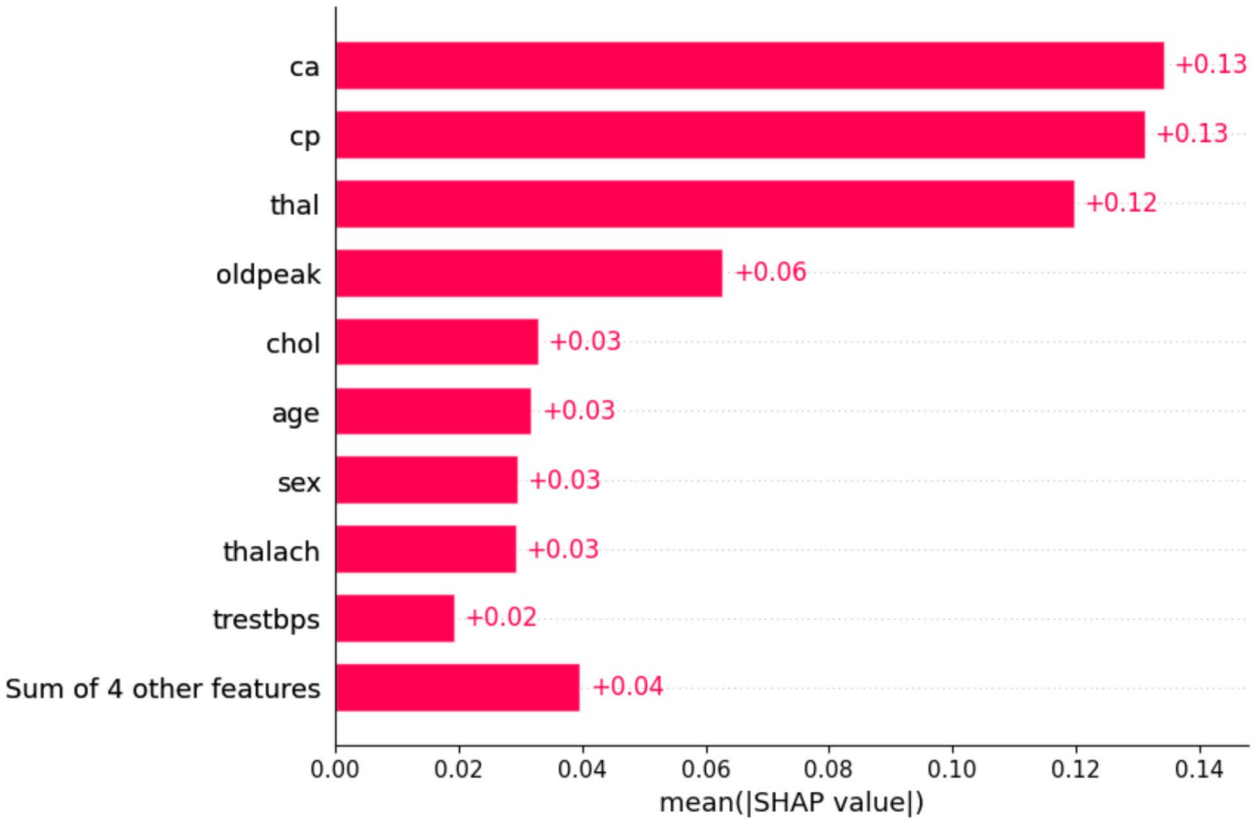


Fig. 5 Mean SHAP values illustrating important features' impact on the RF model in the first approach

Table 6 Performance of models in the second approach

Model	Local model	Accuracy (%)	PPV (%)	NPV (%)	Specificity (%)	Sensitivity or Recall (%)	F1-Score (%)
DT	Local model 1	81.25	78.94	84.61	73.33	88.23	83.33
	Local model 2	74.60	64.86	88.46	63.88	88.88	75.00
	Local model 3	68.25	80.95	61.90	86.66	51.51	62.96
	Local model 4	87.30	82.92	95.45	75.00	97.14	89.47
	Local model 5	76.19	71.87	80.64	73.52	79.31	75.41
	Average	77.51	75.91	82.21	74.48	81.01	77.23
RF	Local model 1	80.70	83.33	77.77	80.76	80.64	81.96
	Local model 2	85.96	81.81	91.66	78.57	93.10	87.09
	Local model 3	85.96	86.11	85.71	78.26	91.17	88.57
	Local model 4	89.47	89.18	90.00	81.81	94.28	91.66
	Local model 5	80.35	80.00	80.76	77.77	82.75	81.35
	Average	**84.49**	**84.09**	**85.18**	79.43	**88.39**	**86.13**
AdaBoost	Local model 1	89.06	88.57	89.65	86.66	91.17	89.85
	Local model 2	77.77	68.57	89.28	69.44	88.88	77.41
	Local model 3	74.60	79.31	70.58	80.00	69.69	74.19
	Local model 4	88.88	83.33	100	75.00	100	90.90
	Local model 5	82.54	87.50	79.48	91.17	72.41	79.24
	Average	82.57	81.45	85.80	**80.45**	84.43	82.32
XGBoost	Local model 1	78.94	80.64	76.92	76.92	80.64	80.64
	Local model 2	85.96	81.81	91.66	78.57	93.10	87.09
	Local model 3	85.96	90.62	80.00	86.95	85.29	87.87
	Local model 4	89.47	89.18	90.00	81.81	94.28	91.66
	Local model 5	80.03	80.00	80.76	77.77	82.75	81.35
	Average	84.14	84.45	83.87	80.40	87.21	85.72
KNN	Local model 1	84.37	85.29	83.33	83.33	85.29	85.29
	Local model 2	76.19	68.75	83.87	72.22	81.48	74.57
	Local model 3	84.12	89.65	79.41	90.00	78.78	83.87
	Local model 4	87.30	84.61	91.66	78.57	94.28	89.18
	Local model 5	76.19	70.58	82.75	70.58	82.75	76.19
	Average	81.63	78.94	84.20	78.94	84.52	81.82

privacy and slightly increases the computational burdens associated with healthcare predictions. By using the IBM AI Privacy Toolkit, the FMLCA demonstrates a commitment to incorporating state-of-the-art privacy-preserving techniques into its method. This toolkit is instrumental in safeguarding sensitive patient data throughout the FL process, aligning with the ethical considerations and privacy requirements inherent in healthcare applications. The successful integration of the IBM AI Privacy Toolkit within FMLCA not only ensures robust privacy measures but also contributes to the overall efficiency of this approach. This implementation choice strengthens the FL's credibility and positions it as a reliable solution for healthcare applications with privacy considerations.

Beyond emphasizing the benefits of ensuring data privacy and security, it is essential to critically examine both the strengths and limitations of privacy-preserving approaches, such as the one utilized in this study. One of the main strengths is the ability to safeguard sensitive patient data during the FL process. By using advanced techniques, such as k-anonymity, the proposed approach ensures data confidentiality and mitigates the risk of unauthorized access. Specifically, k-anonymity was selected in this study due to its efficiency and suitability for the dataset size, effectively reducing the risk of re-identification while maintaining model performance. In comparison, the differential privacy method provides formal privacy guarantees by adding controlled noise to model updates, which can slightly reduce accuracy, whereas the secure multiparty computation method enables computations on encrypted data but introduces considerable computational overhead. Furthermore, the use of the IBM AI Privacy Toolkit ensures compliance with ethical

Table 7 Computational times of models in the second approach

Model	Local model	Time (second)
DT	Local model 1	0.007
	Local model 2	0.003
	Local model 3	0.002
	Local model 4	0.003
	Local model 5	0.003
	Average	0.003
RF	Local model 1	0.117
	Local model 2	0.118
	Local model 3	0.118
	Local model 4	0.117
	Local model 5	0.116
	Average	0.117
AdaBoost	Local model 1	0.073
	Local model 2	0.064
	Local model 3	0.064
	Local model 4	0.064
	Local model 5	0.062
	Average	0.065
XGBoost	Local model 1	0.119
	Local model 2	0.099
	Local model 3	0.100
	Local model 4	0.103
	Local model 5	0.100
	Average	0.904
KNN	Local model 1	0.007
	Local model 2	0.004
	Local model 3	0.006
	Local model 4	0.004
	Local model 5	0.004
	Average	0.005

and legal standards, fostering trust among patients and stakeholders. Additionally, the implementation of such privacy measures promotes the adoption of FL in healthcare by addressing critical privacy concerns.

Nevertheless, privacy-preserving approaches are not without challenges. The incorporation of privacy-preserving mechanisms increases computational overhead due to the additional steps required, such as encryption, decryption, and noise addition. This can result in higher resource consumption and slower processing times. Another potential drawback is the trade-off between privacy and model performance, as techniques like differential privacy may introduce noise that affects the accuracy and utility of the model. Lastly, the integration of privacy-preserving toolkits adds complexity to the system, requiring specialized expertise and additional development efforts to ensure seamless functionality.

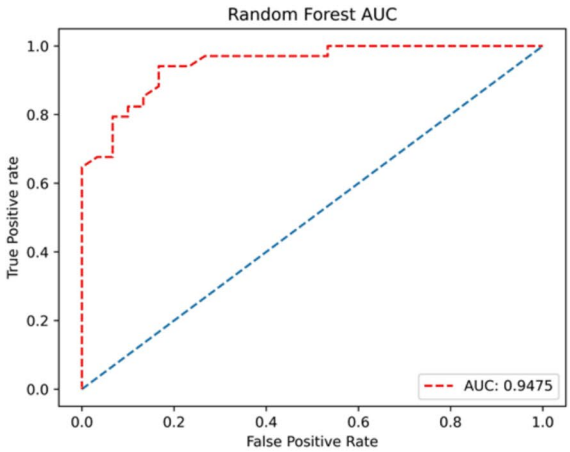
Despite these challenges, the integration of the IBM AI Privacy Toolkit represents a well-considered decision that prioritizes data confidentiality, ethical integrity, and the broader adoption of FL in healthcare. The comparison of different privacy-preserving methods underscores the rationale for selecting k-anonymity in our FMLCA framework, balancing privacy protection and computational efficiency for practical clinical application. These trade-offs are justified given the critical need for robust privacy measures in sensitive domains.

The prominence of heart disease as the leading cause of mortality in the WHO European region has been reiterated in a recent review [55]. Notably, the burden is particularly significant for male patients under the age of 70, with notable variations in patient numbers across different regions [55]. So, there is a compelling need for a high-performance and secure method dedicated to accurate heart disease prediction, with a paramount focus on preserving patient privacy, especially in the European context [55].

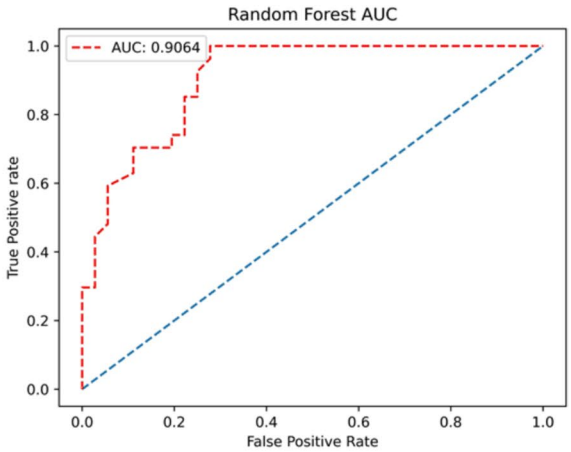
The landscape of healthcare has undergone significant changes due to the Coronavirus Disease 2019 (COVID-19) pandemic, introducing new challenges in disease diagnosis and treatment dynamics [56]. The increased strain on healthcare systems, coupled with the specific effects of COVID-19 on heart patients, underscores the urgency for efficient predictive tools to address this evolving crisis [57, 58]. Furthermore, a recent systematic review has emphasized the crucial role of a treatment decision support system in timely heart disease prediction for patients [59]. This reinforces the importance of developing robust and accurate predictive methods, such as the FMLCA, to aid healthcare professionals in making informed decisions and addressing the evolving landscape of cardiovascular health.

ML models are increasingly used in healthcare for the prediction of disease, detection of patients' smoking behavior, and medication recommendations [56, 60, 61], though their opacity has raised skepticism among clinicians. To address this concern, our study strategically utilized the SHAP technique to augment model transparency. This method aims not only to foster trust within the medical community but also to identify critical predictive features with precision.

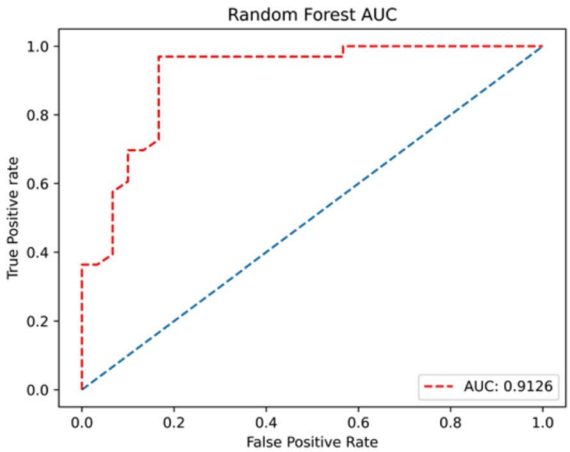
Recent studies also indicate that existing FL and privacy-preserving approaches for heart disease prediction face challenges, including Non-IID data across institutions, communication overhead, and reduced accuracy when noise is introduced by differential privacy mechanisms [62–64]. Our approach, integrating k-anonymity and interpretable modeling, addresses these limitations, enhancing both predictive performance and privacy compliance in clinical settings.



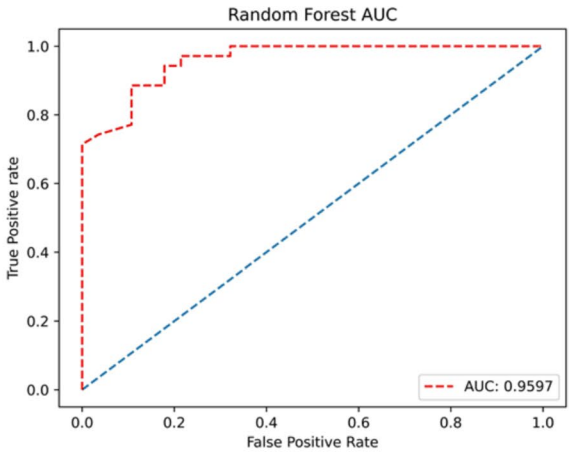
Local model 1



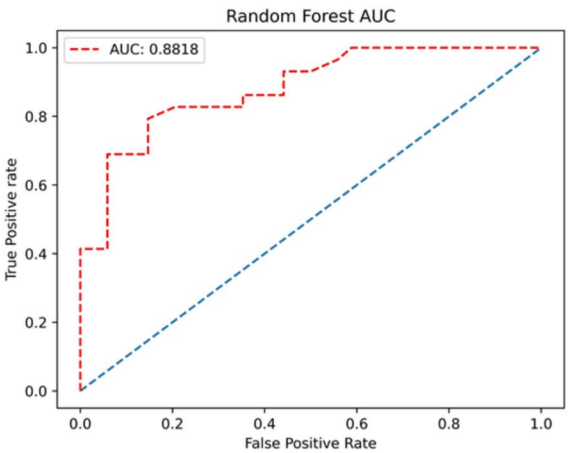
Local model 2



Local model 3



Local model 4



Local model 5

Fig. 6 AUC diagrams illustrating the RF model's performance in the second approach

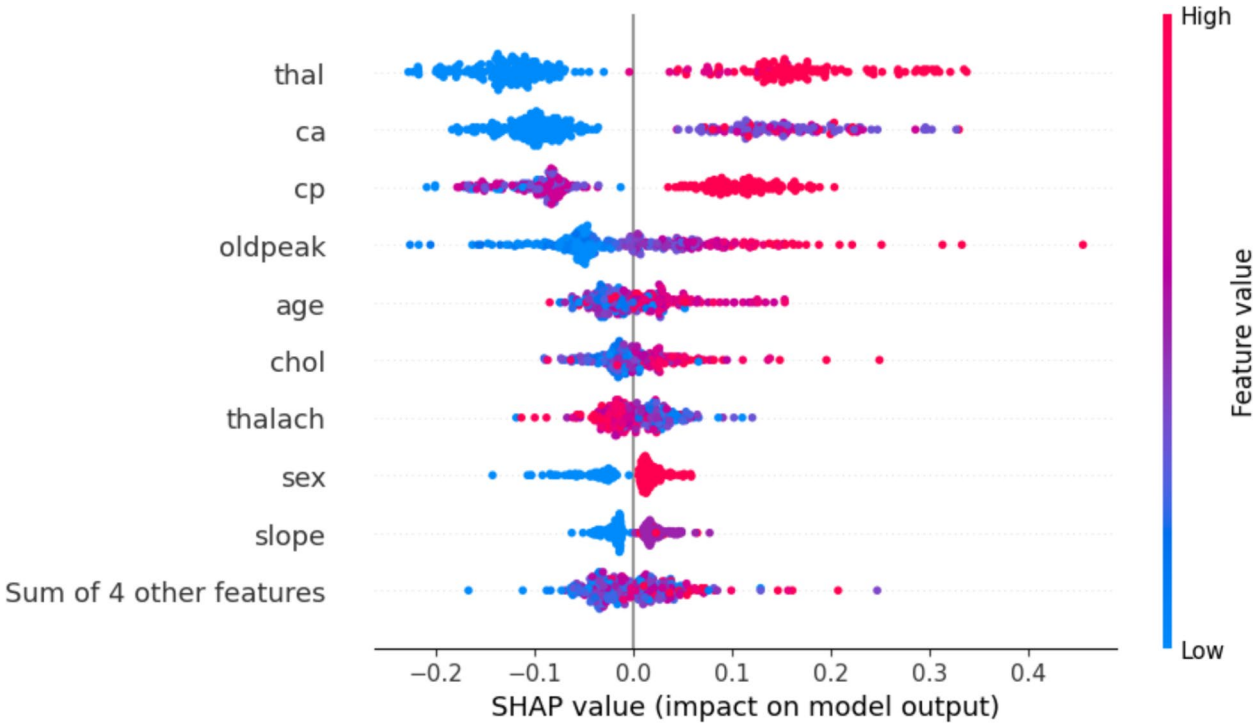


Fig. 7 SHAP values representing the impact of features on the RF model in the second approach

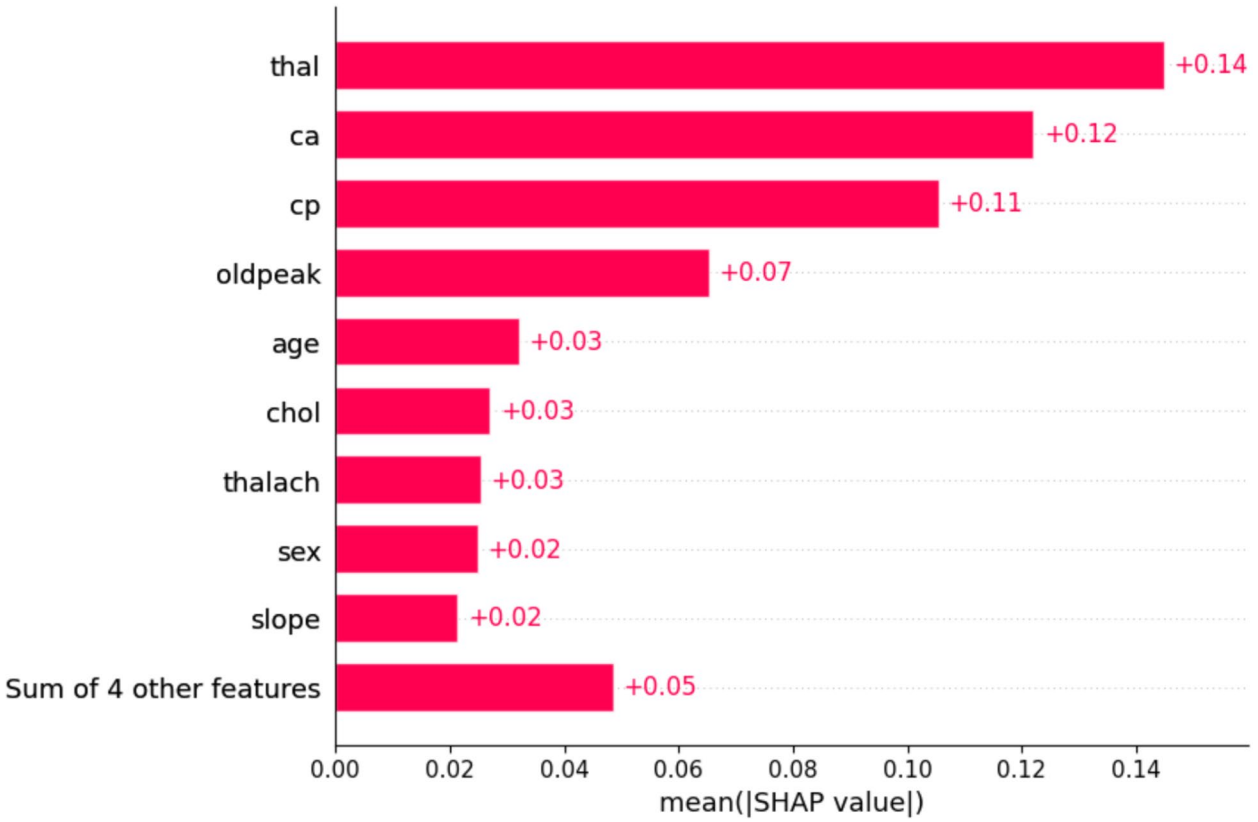


Fig. 8 Mean SHAP values visualizing significant feature influence on the RF model in the second approach

Table 8 Sample t-test results of models in the first approach

	RF	DT	AdaBoost	XGBoost	KNN
RF	-	0.001	0.015	0.011	0.005
DT	0.001	-	0.450	0.890	0.740
AdaBoost	0.015	0.450	-	0.960	0.720
XGBoost	0.011	0.890	0.960	-	0.840
KNN	0.005	0.740	0.720	0.840	-

Table 9 Sample t-test results of models in the second approach

	RF	DT	AdaBoost	XGBoost	KNN
RF	-	0.0001	0.011	0.0003	0.006
DT	0.0001	-	0.104	0.100	0.170
AdaBoost	0.011	0.104	-	0.630	0.620
XGBoost	0.0003	0.100	0.630	-	0.880
KNN	0.006	0.170	0.620	0.880	-

A noteworthy discovery in this study is the persistent significance of features like ca, cp, and thal in the prediction of CAD across both approaches. Using SHAP analysis, these features were identified as the most influential, providing insight into their contributions to the RF model's predictions. This alignment with the emphasis on these features in a prior study [65] underscores their critical role in predictive modeling. Their clinical relevance is evident: for instance, ca reflects the extent of coronary artery narrowing, directly associated with disease severity, while cp and thal capture chest pain type and thalassemia-related markers, respectively, both critical for accurate diagnosis. Correlation analysis further demonstrated the interplay among these features, highlighting how their combined effect enhances model performance. Despite subtle differences in feature ranking between the two approaches, likely due to the integration of privacy-preserving measures in one scenario, these findings reinforce the robustness and interpretability of the proposed FMLCA framework. This detailed feature analysis adds significant clinical and methodological value to the study, supporting informed decision-making for healthcare practitioners.

While this study provides valuable insights into privacy-preserving and explainable FL for CAD prediction, several limitations should be acknowledged. First, the use of a relatively small clinical dataset limits the generalizability of the findings. Although external validation could not be conducted due to the unavailability of additional clinical datasets, the consistent performance of the RF model across local models, evaluation metrics, and federated settings suggests reasonable robustness. Nevertheless, validation on larger, multi-center datasets is required to confirm scalability and broader clinical applicability.

Second, potential sources of bias and confounding may influence the results. Latent factors such as demographic characteristics, comorbidities, or treatment histories may not have been fully captured, and selection bias may arise from the clinical context in which the data were collected. Additionally, missing records were removed rather than imputed to avoid introducing artificial patterns; however, if the missingness was not completely random, this strategy may have introduced bias.

Third, although SMOTE was applied to mitigate class imbalance, synthetic oversampling may affect the underlying data distribution. Alternative imbalance-handling strategies, such as class weighting or cost-sensitive learning, were not explored and could be considered in future studies. Furthermore, computational and resource constraints limited the exploration of a wider hyperparameter search space, which may have prevented achieving optimal model configurations.

Future work will focus on external validation using larger and more diverse clinical datasets, sensitivity and robustness analyses, stratified sampling strategies, and broader hyperparameter optimization. These extensions will further strengthen the generalizability, clinical reliability, and applicability of the proposed FMLCA framework.

Conclusion

Heart disease, as recognized by the WHO, remains a leading cause of mortality worldwide, underscoring the need for reliable and privacy-aware predictive approaches. In this study, multiple ML algorithms were developed and evaluated within a FL framework using a CAD clinical dataset. Data preprocessing included class balancing and the removal of records with non-random missing values to enhance data consistency. Two complementary FL approaches were investigated: one incorporating privacy preservation through k-anonymity and the other without explicit privacy constraints. Across both settings, the RF model demonstrated consistently strong performance relative to the other evaluated algorithms.

The integration of SHAP-based interpretability provided transparent insights into feature contributions, supporting a clearer understanding of model behavior and enhancing trust in the predictive process. While the proposed FMLCA framework demonstrates promising performance and illustrates the feasibility of combining FL, privacy preservation, and explainability for CAD prediction, the findings should be interpreted in light of the dataset size and the absence of external validation.

Overall, the results suggest that the proposed framework may serve as a supportive analytical tool for CAD risk assessment in distributed environments, rather than a standalone diagnostic solution. Future work will focus on validating the framework using larger, multi-center datasets encompassing diverse cardiovascular conditions, as well as further examining clinical integration pathways, contextual applicability, and real-world performance to better assess its practical utility in healthcare settings.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40001-026-04023-6.

Additional file 1.

Author contributions

ASA, FG, LS, AAAS, and JBM were involved in the conception and design of this study. FG preprocessing dataset, and ASA and FG performed the analysis. ASA, LS, AAAS, JBM, and UKW interpreted the results. ASA and FG drafted the manuscript, and all authors (ASA, FG, LS, AAAS, JBM, and UKW) contributed to writing the final draft of the manuscript.

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Availability of data and materials

The dataset analyzed during the current study is available in the "UCI Heart Disease" repository, https://archive.ics.uci.edu/dataset/45.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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